

HTTP (16 Marks)

Diagram – 2 Marks

Definition

HTTP is used to communicate b/w web server and client ..

- An **HTTP client** sends a request.
 - An **HTTP server** returns a response message.
 - The server uses **port number 80**.
 - The client uses a **temporary port number**.
 - HTTP uses the services of **TCP** (connection-oriented).
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Non-Persistent Connections – 3 Marks

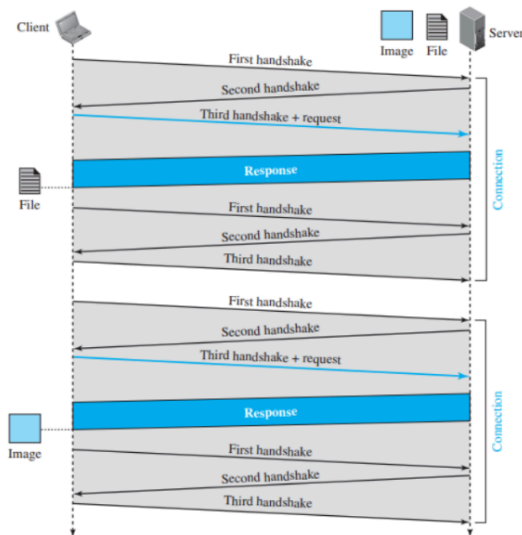
- One **TCP connection** is made for each request/response.

Steps:

1. The client opens a TCP connection and sends a request.
2. The server sends the response and closes the connection.
3. The client reads the data until it encounters an end-of-file (EOF) marker, then closes the connection.

- Nonpersistent connection example

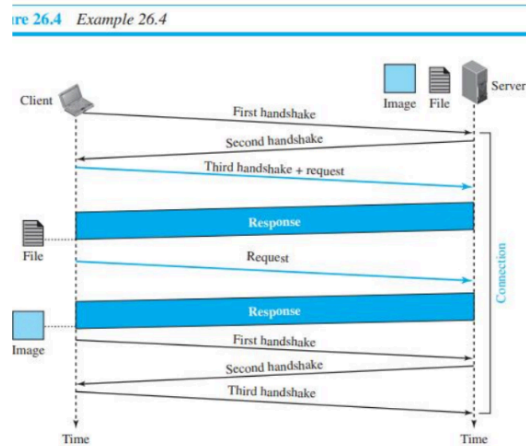
Figure 26.3 Example 26.3



Persistent Connections – 3 Marks

- The server leaves the connection open for **multiple requests** after sending a response.
- The server can close the connection:
 - At the **request of the client**, or
 - If a **timeout** occurs.
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- Persistent Connections Example



Message Format – 2 Marks

Request Message Format:

- Request Line
- Header Lines
- Blank Line
- Body (optional)

Methods: GET , HEAD , PUT , POST , TRACE , DELETE , etc.

Header Fields (Examples):

- User-Agent
- Accept
- Host
- Date
- Upgrade

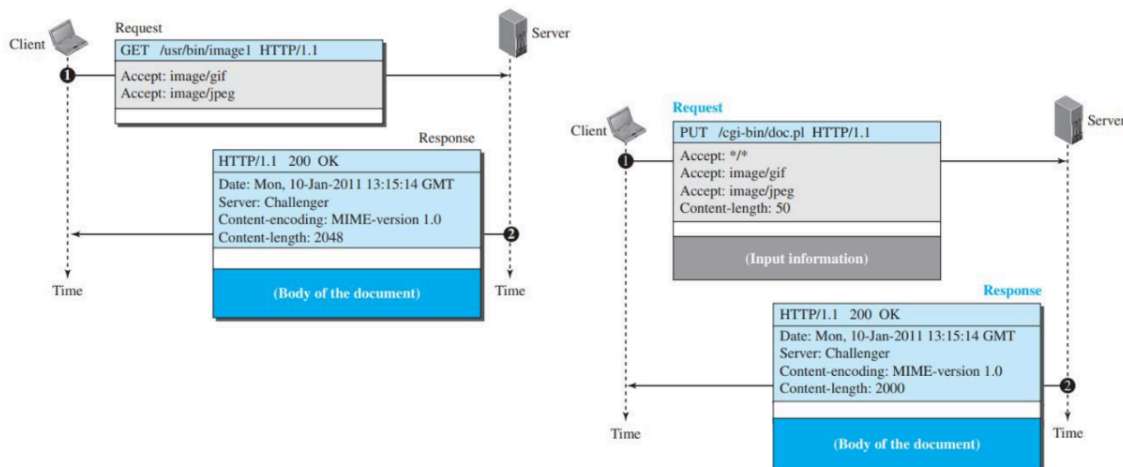
Response Message Format:

- **Status Line**
- **Header Lines**
- **Blank Line**
- **Body** (optional)

Header Fields (Examples):

- **Date**
- **Upgrade**
- **Server**

HTTP- Message Formats



Cookies – 2 Marks

Purpose:

- Websites use cookies as **electronic storage**.
- Some websites allow access only to **registered clients**.
- Cookies are used to manage **portal-based sessions**.

Creating & Storing Cookies – 2 Marks

- When a server receives a request from a client, it stores information like:
 - Domain name
 - Cookie content
 - Timestamp
- The server includes the cookie in the **response** to the client.
- The client stores the cookie in the **browser's cookie directory**, sorted by domain name.

Using Cookies – 2 Marks

- When the client sends a future request to the server:
 - The browser checks the cookie directory.
 - If a matching cookie is found, it's included in the request.
 - The server identifies the client as a **returning user**, not a new one.e) is used instead for secure communication.
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